edunet foundation

Module 3

Machine Learning

Units for Discussion

Chapter - 4	Introduction to Machine Learning
Chapter - 5	Supervised Learning
Chapter - 6	Unsupervised Learning

Chapter

4

Introduction to Machine Learning

DISCLAIMER

The content is curated from online/offline resources and used for educational purpose only.

Learning Objectives

- Understand the historical development and current trends in AI and ML, including their impact across various industries.
- Explain the significance of AI and ML in promoting sustainability and addressing environmental challenges.
- Differentiate between the main types of machine learning: supervised, unsupervised, semi-supervised, and reinforcement learning.
- Outline the essential steps in a machine learning workflow, including data collection, preprocessing, model selection, training, evaluation, and deployment.
- Import and implement basic ML models using Scikit-Learn, including linear regression, decision trees, and clustering algorithms.

Overview of AI and ML

Artificial Intelligence (AI) is the field of computer science focused on creating systems capable of performing tasks that typically require human intelligence. It enables machines to understand, reason, learn, and respond intelligently to various types of inputs. AI systems can simulate human-like cognitive functions such as perception, decision-making, problem-solving, and even creativity

Overview of AI and ML

Type of AI

Artificial Intelligence Types,

1. Narrow AI: AI designed to complete very specific actions; unable to independently learn.
2. Artificial General Intelligence: AI designed to learn, think and perform at similar levels to humans.
3. Artificial Superintelligence: AI able to surpass the knowledge and capabilities of humans.

Overview of AI and ML

Functionality-Based Types of Artificial Intelligence

Functionality based AI types,

1. Reactive Machine AI: AI capable of responding to external stimuli in real time; unable to build memory or store information for future.
2. Limited Memory AI: AI that can store knowledge and use it to learn and train for future tasks.
3. Theory of Mind AI: AI that can sense and respond to human emotions, plus perform the tasks of limited memory machines.
4. Self-Aware AI: AI that can recognize others' emotions, plus has sense of self and human-level intelligence; the final stage of AI.

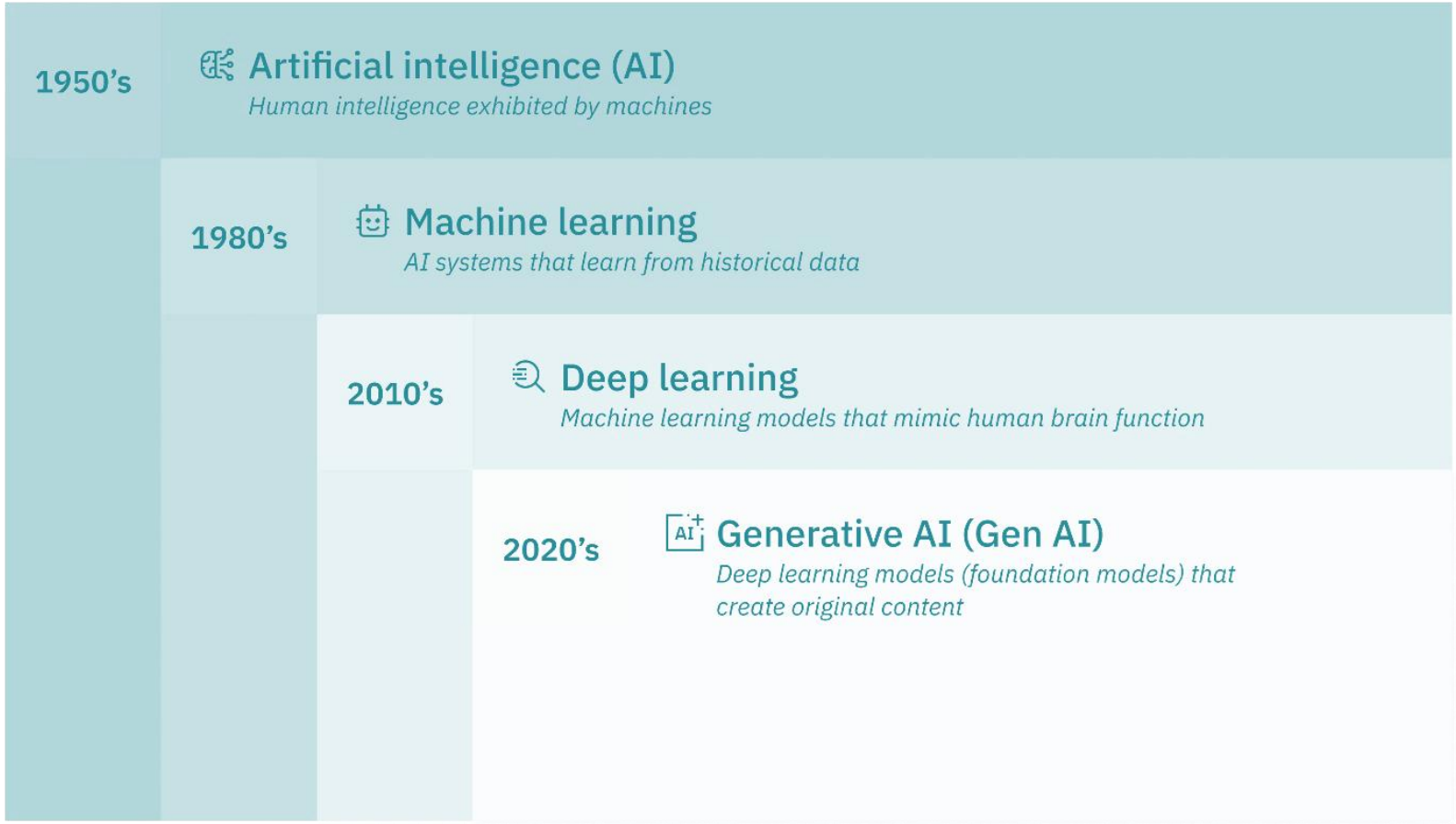
Overview of AI and ML

AI Concepts

1. Machine Learning (ML) and Deep Learning (DL)
2. Natural Language Processing (NLP)
3. Computer Vision
4. Reinforcement Learning (RL)
5. Generative Models
6. Explainable AI (XAI)

Overview of AI and ML

History of AI



Overview of AI and ML

Application of AI

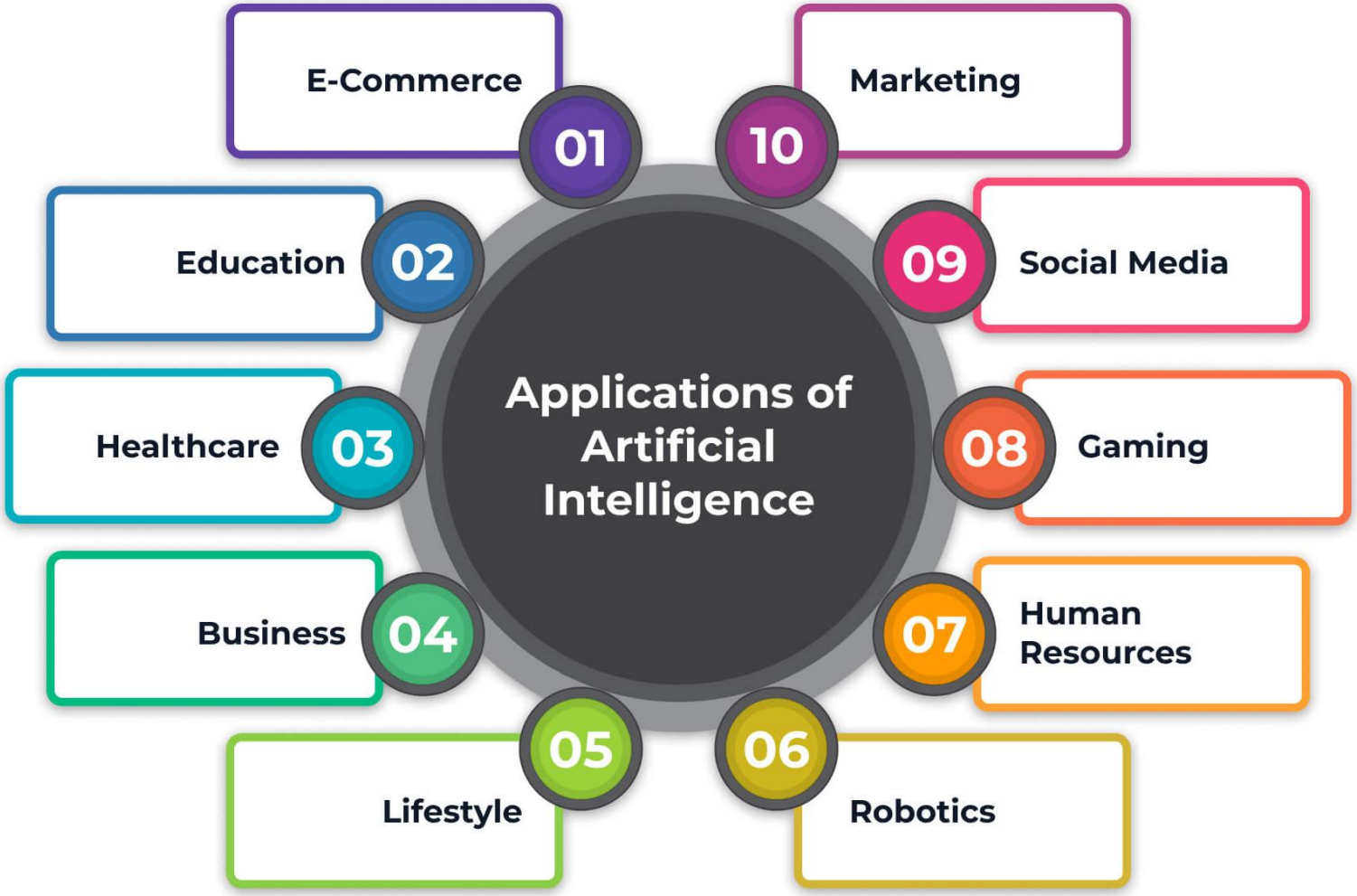
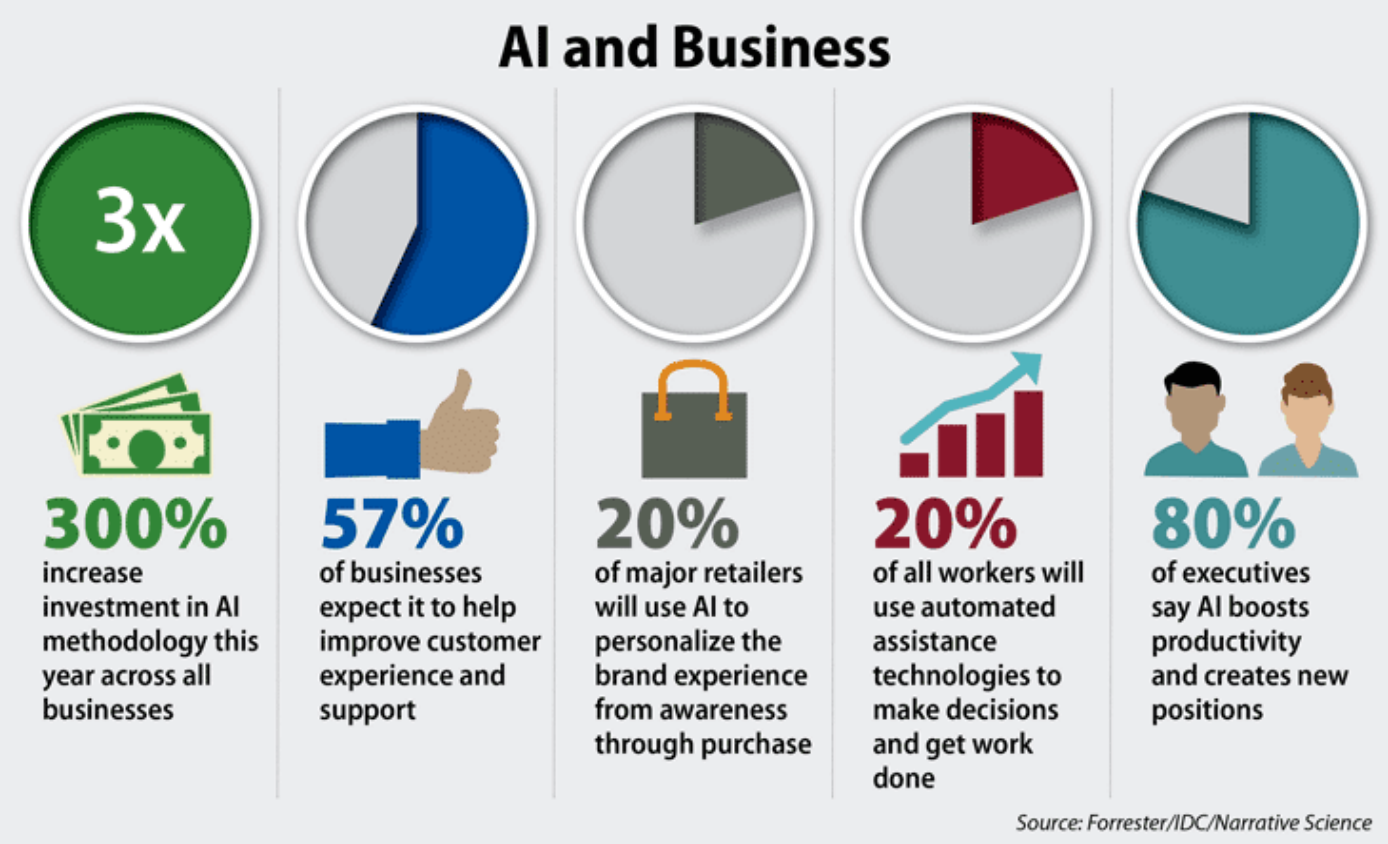


Image source : https://www.solulab.com/top-artificial-intelligence-applications/

Overview of AI and ML

Jobs Opportunities of AI-ML



Diverse Job Roles

- Data Scientist
- Machine Learning Engineer
- AI Research Scientist
- Business Intelligence Analyst
- Data Engineer
- AI Ethics Specialist

Overview of AI and ML

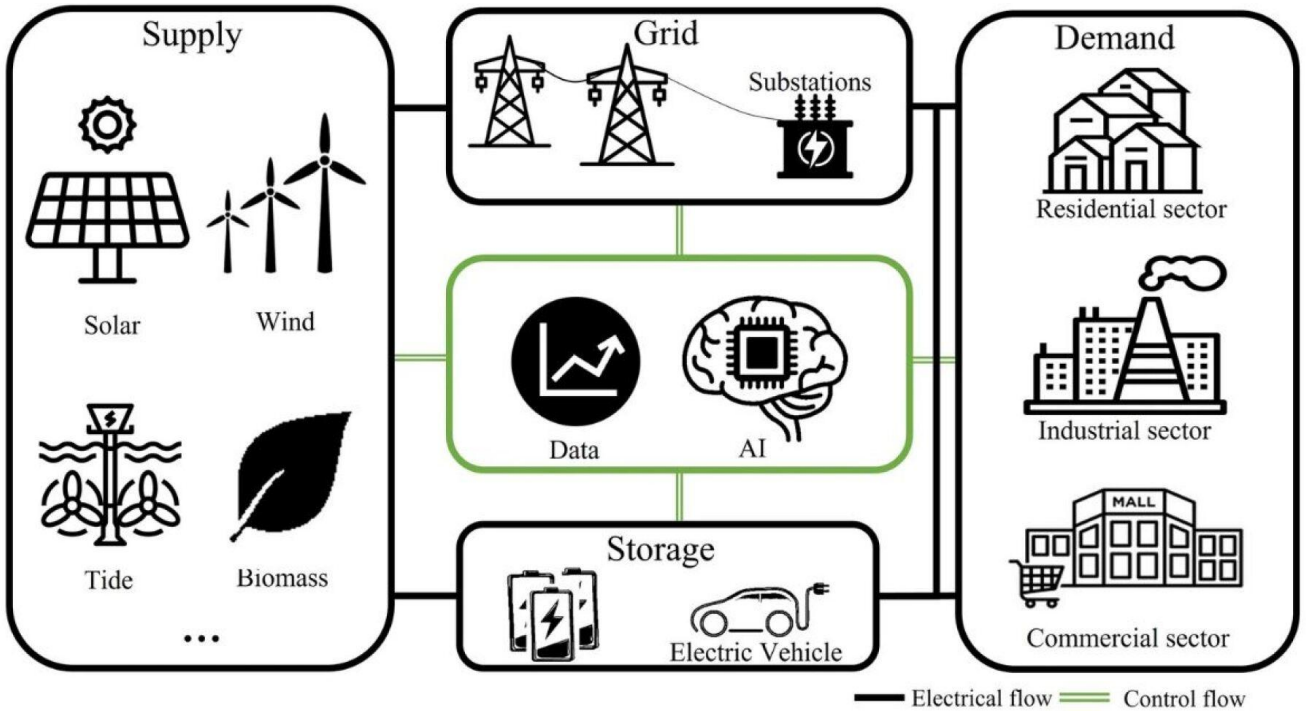
Jobs Opportunities of AI-ML - Skill Requirements

To succeed in AI and ML roles, candidates typically need:

- Proficiency in programming languages (e.g., Python).
- Familiarity with machine learning frameworks (e.g., TensorFlow, Keras, PyTorch) and other basic libraries.
- A solid understanding of statistics, data analysis, and data visualization.

AI-ML Importance of Sustainability

Optimizing Energy Consumption



- **Smart Grids**: AI can predict energy demand and adjust electricity distribution, reducing waste and ensuring that power is available where it's needed most.
- **Energy Efficiency**: ML models analyze how buildings use energy and suggest ways to reduce consumption, such as optimizing heating and cooling systems to lower carbon emissions.
- **Renewable Energy**: AI helps renewable energy systems, like solar panels and wind turbines, by predicting energy production and making sure it’s stored efficiently for when it’s needed.

Image source : https://arxiv.org/html/2406.15732v1

AI-ML Importance of Sustainability

Improving Agricultural Practices

- **Precision Agriculture**: Farmers can use data from drones, sensors, and satellites to monitor soil health, water usage, and crop growth, allowing them to use fewer resources while still getting great results.
- **Yield Prediction**: ML can predict how much food crops will produce, helping farmers plan better and avoid overusing water, fertilizer, and pesticides.
- **Sustainable Farming**: AI helps farmers reduce waste and improve efficiency by optimizing planting schedules and making supply chains more sustainable.

AI-ML Importance of Sustainability

Supporting Climate Change Mitigation

- **Climate Modeling**: These technologies process vast amounts of data to improve weather and climate predictions, helping us understand how the climate is changing and what we can do to mitigate its impact.
- **Carbon Footprint Reduction**: By optimizing industrial processes, transportation routes, and energy consumption, ML can help industries reduce their carbon emissions.
- **Disaster Management**: AI can predict natural disasters like floods and wildfires, giving people more time to prepare and minimize environmental and human damage.

AI-ML Importance of Sustainability

Enhancing Resource Management

- **Water Management**: AI systems can monitor water quality, predict future demand, and detect leaks in water supply systems, making sure we use this vital resource wisely.
- **Waste Reduction**: ML models optimize recycling processes and promote circular economies, where products are reused or recycled instead of ending up in landfills.
- **Sustainable Supply Chains**: AI makes supply chains more sustainable by reducing energy consumption and waste during transportation and production, while ensuring materials are responsibly sourced.

AI-ML Importance of Sustainability

Promoting Sustainable Urban Development

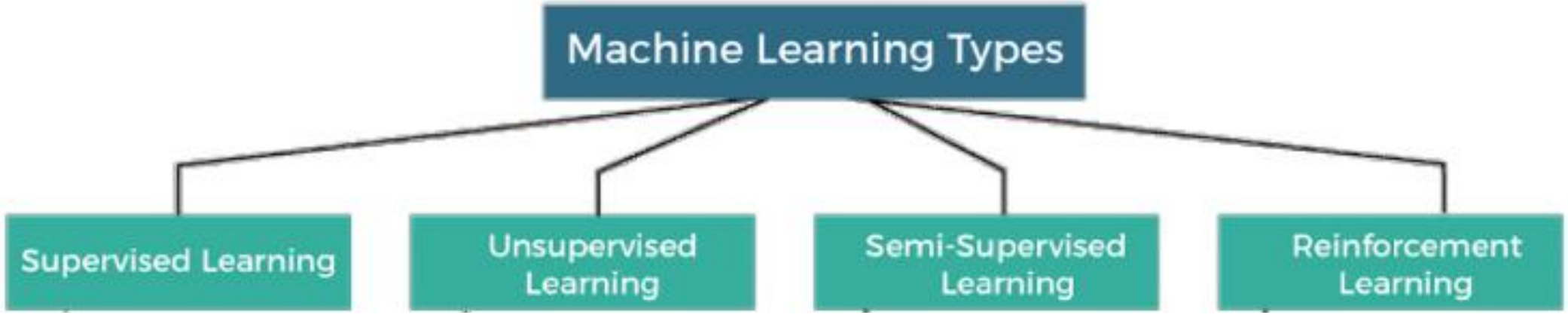
- **Smart Cities**: AI-powered systems can optimize traffic, reduce congestion, and improve public transportation, all of which contribute to better air quality and reduced energy consumption.
- **Sustainable Infrastructure**: ML helps design energy-efficient buildings, reducing waste and ensuring urban environments are more environmentally friendly.
- **Urban Resilience**: By predicting how climate change will affect cities, AI helps urban planners build more resilient and sustainable communities.

AI-ML Importance of Sustainability

Advancing Environmental Conservation

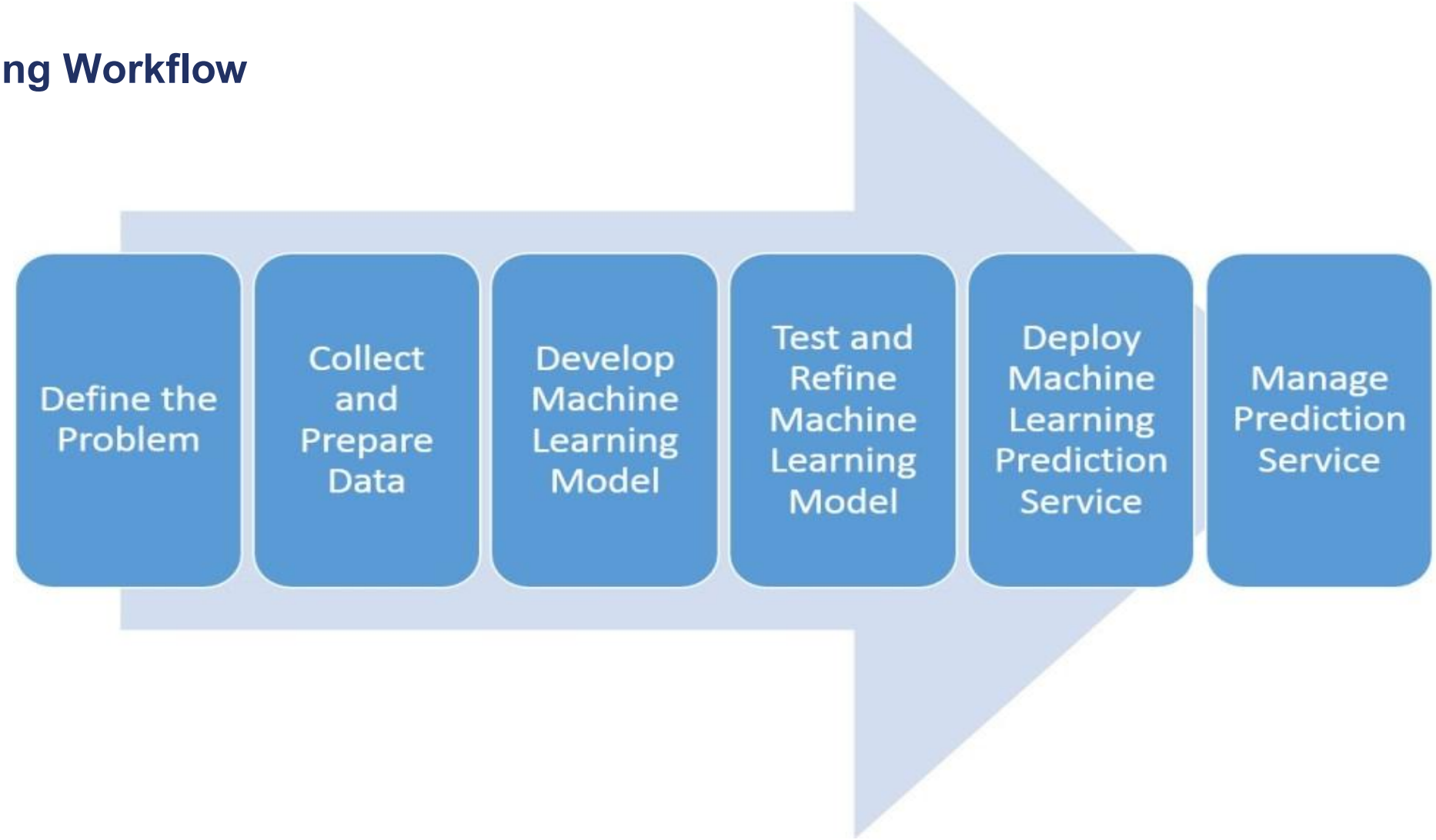
- **Wildlife Monitoring**: AI systems, using drones and sensors, help track endangered species, protect wildlife from poaching, and monitor animal populations in remote areas.
- **Forest Management**: Machine learning analyzes satellite data to detect deforestation, assess forest health, and guide reforestation efforts, ensuring that forests remain healthy and productive.
- **Biodiversity Protection**: AI helps assess the impact of human activity on ecosystems, providing the insights needed to protect biodiversity and maintain balanced ecosystems.

Types of Machine Learning



Within the field of artificial intelligence, machine learning uses statistical models and algorithms to find patterns in data and generate predictions without the need for explicit programming.

Machine Learning Workflow



Machine Learning Python Package (Scikit-Learn)

Installation and Setup

You can check your Python and pip versions by running

```
PS C:\Users\RAMAR BOSE> python --version
Python 3.12.4
PS C:\Users\RAMAR BOSE> pip --version
pip 24.0 from C:\Users\RAMAR BOSE\anaconda3\Lib\site-packages\pip (python 3.12)
PS C:\Users\RAMAR BOSE>
```

To install scikit-learn, open a terminal or command prompt / anaconda prompt or juypter notebook and type

```
PS C:\Users\RAMAR BOSE> pip install scikit-learn
Requirement already satisfied: scikit-learn in c:\users\ramar bose\anaconda3\lib\site-packages (1.3.2)
Requirement already satisfied: numpy<2.0,>=1.17.3 in c:\users\ramar bose\anaconda3\lib\site-packages (from scikit-learn) (1.26.3)
Requirement already satisfied: scipy>=1.5.0 in c:\users\ramar bose\anaconda3\lib\site-packages (from scikit-learn) (1.13.1)
Requirement already satisfied: joblib>=1.1.1 in c:\users\ramar bose\anaconda3\lib\site-packages (from scikit-learn) (1.4.2)
Requirement already satisfied: threadpoolctl>=2.0.0 in c:\users\ramar bose\anaconda3\lib\site-packages (from scikit-learn) (3.5.0)
```

Machine Learning Python Package (Scikit-Learn)

Data Preparation

```
from sklearn.datasets import l
```

```
load_breast_cancer
load_diabetes
load_digits
load_files
load_iris
load_linnerud
load_sample_image
load_sample_images
load_svmlight_file
load_svmlight_files
load_wine
{} _lfw
```

```
from sklearn.datasets import load_iris
# Load the Iris dataset
data = load_iris()
# Features and target
X, y = data.data, data.target
```

✓ 0.0s

Machine Learning Python Package (Scikit-Learn)

Data Splitting

```
from sklearn.model_selection import train_test_split

# Assume X is your feature matrix and y is your target vector
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```
✓ 0.1s

Parameters:

- X: The feature matrix to be split.
- y: The target vector to be split.
- test_size=0.2: This parameter specifies the proportion of the dataset to include in the test split. In this case, 20% of the data will be reserved for testing, and the remaining 80% will be used for training.
- random_state=42: This parameter is used for reproducibility. By providing a specific integer (like 42), you ensure that every time you run your code, you will get the same split of data. If you do not set a random_state, the split will be different each time you run the code, making your results hard to reproduce.

Machine Learning Python Package (Scikit-Learn)

Data Preprocessing

Feature Scaling

Standardization (Z-score normalization)

```python
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```
✓ 0.0s

Min-Max Scaling

```python
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
X_scaled = scaler.fit_transform(X)
```
✓ 0.0s

Handling Missing Values

```python
from sklearn.impute import SimpleImputer

# Create an imputer object with a strategy
imputer = SimpleImputer(strategy='mean')

# Fit the imputer on the data and transform it
X_imputed = imputer.fit_transform(X)
```
✓ 0.9s

Encoding Categorical Variables

```python
from sklearn.preprocessing import OneHotEncoder

encoder = OneHotEncoder()
X_encoded = encoder.fit_transform(X_categorical)
```

Label Encoding for ordinal variables

```python
from sklearn.preprocessing import LabelEncoder

label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y_categorical)
```

Machine Learning Python Package (Scikit-Learn)

Model Selection

```python
from sklearn.linear_model import LogisticRegression

# Instantiate the model
model = LogisticRegression()
```

Training the Model

```python
# Fit the model on the training data
model.fit(X_train_scaled, y_train)
```

Model Evaluation

```python
from sklearn.metrics import
```

f	accuracy_score	function
f	adjusted_mutual_info_score	function
f	adjusted_rand_score	function
f	auc	function
f	average_precision_score	function
f	balanced_accuracy_score	function
f	brier_score_loss	function
f	calinski_harabasz_score	function
f	check_scoring	function
f	class_likelihood_ratios	function

Machine Learning Python Package (Scikit-Learn)

Hyperparameter Tuning

```
param_grid = {
    'n_estimators': [50, 100, 200],          # Number of trees
    'max_depth': [None, 10, 20, 30],        # Maximum depth of the tree
    'min_samples_split': [2, 5, 10],        # Minimum samples required to split an internal node
}
```

```
from sklearn.model_selection import GridSearchCV

# Instantiate the GridSearchCV object
grid_search = GridSearchCV(estimator=model, param_grid=param_grid,
                           cv=5, scoring='accuracy', n_jobs=-1)
# Fit the grid search to the training data
grid_search.fit(X_train, y_train)
# View the best hyperparameters
print(f"Best Hyperparameters: {grid_search.best_params_}")
```

Machine Learning Python Package (Scikit-Learn)

Model Saving and Deployment

```
import joblib
# Save the model
joblib.dump(model, 'random_forest_model.joblib')
```

```
import pickle

# Save the model
with open('random_forest_model.pkl', 'wb') as file:
    pickle.dump(model, file)
```

Conclusion

- Learned the historical development and current trends in AI and ML, including their impact across various industries.
- Explained the significance of AI and ML in promoting sustainability and addressing environmental challenges.
- Understood the differentiate between the main types of machine learning: supervised, unsupervised, semi-supervised, and reinforcement learning.
- Understood the outline the essential steps in a machine learning workflow, including data collection, preprocessing, model selection, training, evaluation, and deployment.
- Explored and imported and implemented the basic ML models using Scikit-Learn, including linear regression, decision trees, and clustering algorithms.

Reference

1. Bishop, C. M. (2006). *Pattern Recognition and Machine Learning.* Springer.
2. Aggarwal, C. C., & Zhai, C. (2012). *Mining Text Data.* Springer.
3. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning.* MIT Press.
4. Hartigan, J. A., & Wong, M. A. (1979). "Algorithm AS 136: A k-means clustering algorithm." *Journal of the Royal Statistical Society, Series C (Applied Statistics).*
5. Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning.* Springer.
6. Jain, A. K., Murty, M. N., & Flynn, P. J. (1999). "Data clustering: A review." *ACM Computing Surveys, 31(3),* 264–323.
7. Friedman, J., Hastie, T., & Tibshirani, R. (2001). *The Elements of Statistical Learning.* Springer Series in Statistics.

Q&A
Let’s Start

Quiz

1. **Which of the following best describes Artificial Intelligence (AI)?**

a) The simulation of human intelligence by computers

b) The study of electronic systems and circuits

c) A branch of physics focused on matter and energy

d) The programming language used for mobile applications

Answer: a

The simulation of human intelligence by computers

Quiz

2. How does Machine Learning differ from traditional programming?

a) ML requires more code than traditional programming

b) ML systems improve automatically with experience, whereas traditional programming requires explicit instructions

c) ML does not need any programming at all

d) ML is limited to only numerical data processing

Answer: b

ML systems improve automatically with experience, whereas traditional programming requires explicit instructions

Quiz

3. Which of the following best describes how AI contributes to sustainability?

a) By creating more jobs in manufacturing industries

b) By helping in decision-making processes to minimize environmental impact

c) By increasing energy consumption across all industries

d) By focusing on traditional manufacturing techniques

Answer: b

By helping in decision-making processes to minimize environmental impact

Quiz

4. How does Machine Learning help in predicting environmental changes?

a) By manually categorizing environmental data

b) By analyzing historical data to forecast future environmental patterns

c) By eliminating the need for data in research

d) By focusing only on natural resources without considering human activity

Answer: b

By analyzing historical data to forecast future environmental patterns

Quiz

5. What is the first step in a typical Machine Learning workflow?

a) Model training

b) Data collection

c) Model evaluation

d) Model deployment

Answer: b

Data collection

Thank You